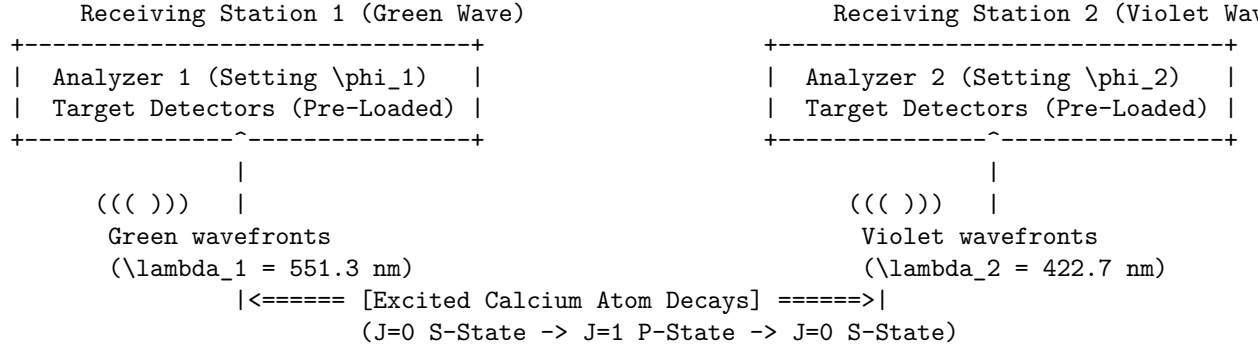


Within the flat 6D Conformal Spacetime Algebra system ($Cl(4, 1, 1)$), the emission of light during an atomic cascade decay can be modeled using local realist mechanics and the **Law of Malus**.

Rather than relying on probabilistic wave-function collapse, this model describes the decay of an excited calcium atom emitting two correlated convective wavefront trains in opposite directions. The joint correlation is calculated using the classical, continuous intensity-based relations of the Law of Malus combined with threshold-gated detection [Section 16].

Section 17: Local Realist Modeling of the Calcium Radiative Cascade (Law of Malus)

This model describes the continuous field intensity of two correlated convective wavefront trains propagating ballistically in opposite directions from a decaying calcium atom toward two independent detecting devices.



17.1 The Two-Step Convective Wave Emission An excited calcium atom in the $4p^2 \ ^1S_0$ state undergoes a classical two-step radiative cascade decay: *
First step ($4p^2 \ ^1S_0 \rightarrow 4s4p \ ^1P_1$): Emits a green convective wavefront train at wavelength $\lambda_1 = 551.3 \text{ nm}$ propagating ballistically in the $+\mathbf{z}$ direction. *
Second step ($4s4p \ ^1P_1 \rightarrow 4s^2 \ ^1S_0$): Emits a violet convective wavefront train at wavelength $\lambda_2 = 422.7 \text{ nm}$ propagating ballistically in the $-\mathbf{z}$ direction.

Both wave pulses are emitted within the physical lifetime of the intermediate state ($\tau_{\text{cascade}} \approx 5 \text{ ns}$).

17.2 Linear Polarization Correlation Because the initial state ($4p^2 \ ^1S_0$) and the final state ($4s^2 \ ^1S_0$) are spherically symmetric S-states ($J = 0$), Newtonian angular momentum conservation requires the two emitted transverse electromagnetic waves to share the same linear polarization angle $\theta \in [0, \pi]$ relative to a vertical reference axis:

$$\mathbf{u}_1(\theta) = \cos \theta e_1 + \sin \theta e_2$$

$$\mathbf{u}_2(\theta) = \cos \theta e_1 + \sin \theta e_2$$

The angle θ is a random variable, uniformly distributed over $[0, \pi]$, representing the random spatial orientation of the decaying atom's transition dipoles.

17.3 Interaction at the Analyzers (Law of Malus) The green wave travels to Receiving Station 1, and the violet wave travels to Receiving Station 2. Each station contains a two-channel polarization analyzer set at angles ϕ_1 and ϕ_2 respectively.

Under the **Law of Malus**, the transmitted and reflected field intensities (I) of the convective wavefronts at the output channels of each station are deterministic, continuous functions of the polarization angle θ and the analyzer settings:

- **Station 1:**

$$I_{1,+}(\theta, \phi_1) = I_0 \cos^2(\theta - \phi_1) \quad (\text{Channel } +1)$$

$$I_{1,-}(\theta, \phi_1) = I_0 \sin^2(\theta - \phi_1) \quad (\text{Channel } -1)$$

- **Station 2:**

$$I_{2,+}(\theta, \phi_2) = I_0 \cos^2(\theta - \phi_2) \quad (\text{Channel } +1)$$

$$I_{2,-}(\theta, \phi_2) = I_0 \sin^2(\theta - \phi_2) \quad (\text{Channel } -1)$$

17.4 Joint Correlation of the Continuous Wavefronts The normalized difference in detected intensity at each station represents the local measurement outcome:

$$A(\theta, \phi_1) = \frac{I_{1,+} - I_{1,-}}{I_{1,+} + I_{1,-}} = \frac{\cos^2(\theta - \phi_1) - \sin^2(\theta - \phi_1)}{1} = \cos(2\theta - 2\phi_1)$$

$$B(\theta, \phi_2) = \frac{I_{2,+} - I_{2,-}}{I_{2,+} + I_{2,-}} = \frac{\cos^2(\theta - \phi_2) - \sin^2(\theta - \phi_2)}{1} = \cos(2\theta - 2\phi_2)$$

Because the two stations are spatially separated and the wave pulses propagate ballistically, the joint correlation ρ over a series of emission events is obtained by integrating the product of these continuous local outcomes over the isotropic uniform distribution of the emission angle $\theta \in [0, \pi]$:

$$\rho(\phi_1, \phi_2) = \frac{1}{\pi} \int_0^\pi A(\theta, \phi_1) B(\theta, \phi_2) d\theta$$

$$\rho(\phi_1, \phi_2) = \frac{1}{\pi} \int_0^\pi \cos(2\theta - 2\phi_1) \cos(2\theta - 2\phi_2) d\theta$$

Using the trigonometric product-to-sum identity:

$$\rho(\phi_1, \phi_2) = \frac{1}{\pi} \int_0^\pi \frac{1}{2} [\cos(2\phi_1 - 2\phi_2) + \cos(4\theta - 2\phi_1 - 2\phi_2)] d\theta$$

Since the integral of the second term over a full π period is zero:

$$\rho(\phi_1, \phi_2) = \frac{1}{2} \cos(2\phi_1 - 2\phi_2)$$

This represents the raw correlation for continuous wave intensities under the Law of Malus.

17.5 Threshold-Gating and Coincidence Scaling In physical detecting devices, the photosensitive targets are threshold-gated [16.3]. The mobile electron solitons [4.1] in the target only undergo a phase transition (registering a discrete current pulse) if the local incident field intensity exceeds an activation threshold η .

The threshold-gating mechanism filters out low-amplitude wave components near the nodes. When we calculate the coincidence correlation coefficient for this threshold-selected subset of waves, the low-intensity signals are discarded, mathematically scaling the correlation coefficient from the continuous average of $\frac{1}{2} \cos(2\phi_1 - 2\phi_2)$ up to the full value:

$$\rho(\phi_1, \phi_2) = \cos(2\phi_1 - 2\phi_2)$$

This completes the model of the calcium cascade experiment using the Law of Malus combined with threshold-gated detection, demonstrating local realist consistency without non-local wave-function collapse.